

NAGA VENKATA SAI CHENNU

AI/ML Engineer | LLM & Agentic Systems | Model Evaluation & Fine-Tuning
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Summary

Experienced AI systems engineer with graduate research experience building large-scale LLM pipelines, extraction services, and evaluation systems. Delivered a Claude API extraction service that converted 500+ filings to validated JSON with 95%+ accuracy and created LLM Forge, a 12-stage fine-tuning pipeline that cut experiment setup time by half. Published peer-reviewed research and filed a patent in AI-driven cybersecurity, with experience in AWS, RAG, and agentic workflows.

TECHNICAL SKILLS

Languages & Tools: Python, JavaScript/TypeScript, Node.js, SQL, Bash, Git, Docker, Linux, Claude Code, Cursor, GitHub Copilot
LLM & Agents: Claude API, OpenAI API, RAG, prompt engineering, function/tool calling, structured outputs, Model Context Protocol (MCP), agentic workflows, Agentic AI Models, Vector Databases, Pydantic v2
Fine-Tuning & ML: LoRA, QLoRA, GRPO, PEFT, TRL, Hugging Face Transformers, PyTorch, scikit-learn, CatBoost, Machine Learning, Deep Learning, distributed training, dataset curation
Evaluation: Automated eval harnesses, benchmark design, VLM evaluation, human-labeled ground truth, telemetry, McNemar’s exact test, ANOVA
Infrastructure: 4x NVIDIA H100 / SLURM HPC clusters, CUDA, AWS, Cloud Services Azure, Data Pipelines For ML, REST APIs, webhooks, OAuth, SQLite, NVIDIA Jetson edge deployment, Healthcare Informatics, Generative AI Model Development

EXPERIENCE

- George Mason University, Costello College of Business

Aug 2025 - May 2026

Graduate Research Assistant, AI Systems Engineering

 - Shipped a Claude API extraction service using generative AI model development techniques to convert 500+ multi-page legal and USPTO filings into schema-validated JSON at 95%+ field accuracy, cutting manual processing effort ~80%; built on a custom MCP server that processed 2,000+ records in production.
 - Architected LLM Forge, a configuration-driven 12-stage deep learning fine-tuning pipeline using LoRA, QLoRA, and GRPO on a 4x NVIDIA H100 SLURM cluster; packaged it for researcher self-service, cutting time-to-experiment 50%+.
 - Built the evaluation and reliability layer required to deploy agentic AI models with confidence: ~900 automated evaluations against human-labeled ground truth, plus telemetry surfacing low-confidence and failed runs before they reached users.
 - Diagnosed a TRL v0.20.0 regression that silently dropped assistant-only loss masking, restoring correct supervised fine-tuning behavior across the training pipeline.
 - Served as primary technical point of contact for two faculty research groups over a 10-month appointment: scoped ambiguous requirements into executable systems, translated model-behavior tradeoffs for non-technical stakeholders, wrote SOPs, and published trained models to Hugging Face.
- KL University, Department of Computer Science & Engineering

Jan 2022 - May 2024

Undergraduate Researcher, Applied ML & AI Systems

 - Built end-to-end Python data pipelines for ML across cybersecurity, NLP, and predictive modeling; combined four machine learning classifiers into an ensemble reaching 93% accuracy and 92% F1, validated with t-tests and ANOVA rather than single-run results.
 - Co-authored five peer-reviewed publications (two first-author, 50+ citations), including at IEEE ICICT 2023, and filed an Indian patent (Smart FuelGuard); deployed a CatBoost model into a web dashboard for non-technical reviewers.

RESEARCH

- Vision-Language Model Visualization Literacy Across 18 Chart Types.2026

PROJECTS

- JetBot - Autonomous Edge AI Agent

 - Built a Telegram-controlled autonomous LLM agent running fully on-device on an NVIDIA Jetson Orin Nano, combining OpenRouter/Nemotron inference, SQLite FTS5 retrieval for conversational memory, and cgroups v2 resource isolation to maintain stable long-running inference on constrained hardware.
- SmartRemit.ai - Conversational Remittance Platform

 - Developed a WhatsApp-native conversational AI workflow for money transfers, handling natural-language intake, recipient and rate resolution, and transaction handoff without relying on a form-based user interface.

EDUCATION & CERTIFICATIONS

- M.S., Computer Science

George Mason University (GPA: 3.57/4.00)

May 2026

Fairfax, VA
- B.Tech, Computer Science & Engineering, Specialization in AI & Intelligent Process Automation

KL University (GPA: 8.7/10)

May 2024

India

 - Achievements:** First Class with Distinction

EDUCATION & CERTIFICATIONS

- AWS Certified AI Practitioner & Cloud Practitioner
- Anthropic Claude Code in Action
- NVIDIA-Certified Associate: Generative AI & LLMs